Investigating Spatial Perception Through an Audio-Based Head-Tracking System Fall 2025 Thesis Progress Report

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Audio-augmented reality (AAR) systems offer a promising pathway for enhancing spatial awareness through sound, particularly for blind and low-vision users who rely on non-visual cues for navigation. When combined with accurate head tracking, spatial audio allows sound sources to remain perceptually anchored to fixed locations in the environment as a user moves, supporting intuitive understanding of spatial layout.

This progress report documents the Fall 2025 development toward a head-tracked spatial audio system using low-cost inertial sensing and real-time audio processing. The semester focused on establishing a robust spatial audio engine and integrating an IMU-based head-tracking pipeline capable of supporting world-locked audio behavior. The report summarizes completed work, validation to date, challenges encountered, and planned next steps.

1. Semester Progress Overview

The focus was on establishing the core technical foundation required for a real-time audio-augmented reality system. Progress during the semester can be summarized as follows.

- Implementation of a real-time HRTF-based spatial audio engine that supports multiple simultaneous sound sources.
- Development of smoothing and interpolation strategies to eliminate audible artifacts during dynamic source movement.
- Integration of a 6-DOF IMU-based head-tracking system using a Madgwick sensor fusion algorithm.
- Real-time streaming of head orientation data from a wearable IMU to a host audio system using UDP.
- Validation of world-locked audio behavior using both visualization and auditory feedback.

2. Methods

This project was developed through iterative prototyping of a real-time spatial audio pipeline with head tracking. The early versions of the system prioritized functional correctness, followed by successive refinements that targeted perceptual stability, audio quality, and latency reduction. The final system integrates inertial sensing, real-time sensor fusion, network streaming, and HRTF-based spatial audio rendering.

A. System Architecture Overview. The system consists of three primary components:

1. A wearable IMU unit that estimates head orientation in real time.
2. A host-side spatial audio engine that renders world-locked audio using HRTFs.
3. A communication layer that streams orientation data with minimal latency.

Orientation is estimated on the IMU platform and transmitted via UDP to the host machine, where it is used to update the azimuth and elevation parameters of the audio source in real time.

B. IMU Sensing and Orientation Estimation. Head orientation is estimated using an MPU6050 inertial measurement unit providing tri-axial accelerometer and gyroscope measurements. A 6-DOF Madgwick sensor fusion algorithm is used to compute the orientation in quaternion form. Gyroscope measurements provide short-term responsiveness, whereas accelerometer measurements constrain long-term drift in roll and pitch.

The filter gain parameter β was empirically tuned to balance responsiveness and stability. Orientation quaternions are normalized before transmission to avoid numerical drift.

C. Validation: Head-Tracking Consistency Using a 3D Visualizer. Prior to integrating head orientation data into the spatial audio rendering pipeline, a dedicated 3D visualizer was used to validate the correctness of head-tracking behavior. The goal of

Significance Statement

Blind and low-vision individuals rely heavily on spatial hearing to support navigation and environmental awareness, yet affordable technologies that leverage spatial audio remain limited. This project investigates the feasibility of a low-cost, wearable audio-augmented reality system that uses real-time head tracking to maintain world-locked spatial audio cues as a user moves. By combining commodity inertial sensors, real-time sensor fusion, and head-related transfer function–based audio rendering, this work explores key technical tradeoffs between latency, stability, and perceptual clarity. The long-term goal is to inform the design of accessible spatial audio interfaces for assistive navigation and inclusive human–computer interaction.

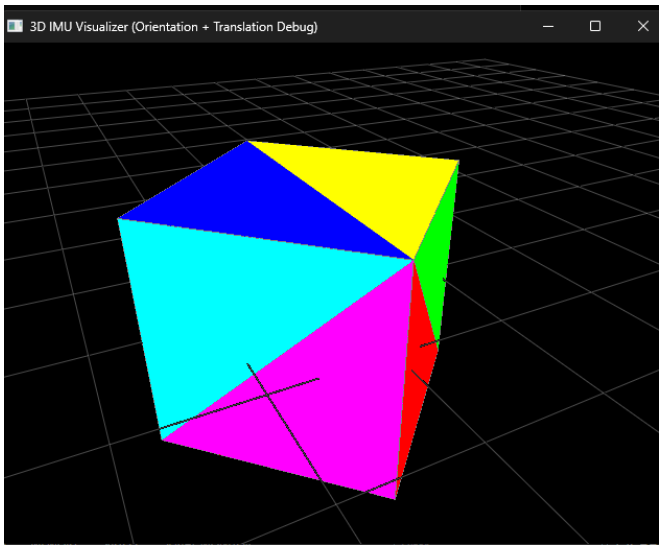

Fig. 1. 3D head-tracking visualizer used for kinematic validation. A rigid cube represents head orientation and is updated in real time using incoming quaternion data streamed from the IMU. Controlled head rotations allow verification of axis alignment, sign conventions, and temporal consistency prior to audio integration.

this validation step was to ensure that inertial measurements, sensor fusion, network transmission, and coordinate frame mappings produced physically consistent and interpretable orientation updates before perceptual audio evaluation.

C.1. Validation Setup. Orientation estimates are computed on the IMU platform using a Madgwick sensor fusion algorithm and transmitted via UDP as quaternions at approximately 100 Hz. Each packet includes a transmission timestamp and quaternion components (q_w, q_x, q_y, q_z).

On the host machine, incoming quaternion data are received asynchronously and applied to a rigid cube rendered in a PyQtGraph-based 3D visualization environment. The quaternion is converted to a rotation matrix and applied directly to the cube transform, allowing head orientation to be observed visually in real time.

C.2. Axis and Sign Convention Verification. The visualizer was used to verify axis alignment and sign conventions between the IMU reference frame and the host visualization frame. Controlled head rotations were performed along individual axes (yaw, pitch, roll), and the resulting cube motion was observed.

In particular, yaw rotation was validated by confirming that a leftward head rotation produced a corresponding leftward rotation of the cube about the vertical axis. An inversion of the z-component of the quaternion was required to reconcile differences between the IMU coordinate frame and the visualization coordinate frame. This correction ensured consistent rotational behavior across both systems.

C.3. Temporal Consistency and Update Rate Validation. Temporal consistency was validated by observing smooth cube motion during continuous head movement without visible jitter, discontinuities, or dropped updates. Orientation updates were received and applied at a rate consistent with the IMU update frequency ($\sim$100 Hz), confirming that UDP-based streaming and non-blocking reception were sufficient for real-time visualization.

Latency measurements were logged concurrently within the visualizer to confirm that orientation updates were applied promptly after transmission. While absolute latency values were affected by unsynchronized clocks, relative timing consistency was sufficient for validating correct kinematic behavior.

C.4. Role of the Visualizer in System Validation. The 3D visualizer served as a critical intermediate validation tool, enabling isolation of head-tracking correctness from audio rendering effects. By confirming that head movements produced accurate and stable visual rotations, errors related to sensor fusion, quaternion normalization, axis inversion, and network transmission were identified and resolved prior to audio integration.

This visual validation established confidence that subsequent spatial audio behavior was driven by audio processing design choices rather than upstream head-tracking errors.

D. Network Streaming and Timing. Orientation estimates are streamed from the IMU platform to the host system using UDP. UDP was selected over TCP to minimize latency and avoid blocking behavior. Each packet includes a transmission timestamp embedded at the IMU, enabling downstream latency analysis. Because system clocks are not synchronized across devices, absolute latency measurements exhibits offset errors as seen in Table 1; therefore, relative latency and jitter are emphasized during evaluation.

E. Spatial Audio Rendering. Spatial audio is rendered using head-related transfer functions (HRTFs) loaded from a SOFA dataset. Audio is processed in fixed-size blocks using overlap-add convolution to enable real-time playback. Each virtual sound source is represented as a mono audio stream convolved independently for the left and right ears.

Early implementations used nearest-neighbor HRTF selection based solely on instantaneous azimuth and elevation. While computationally simple, this approach produced audible artifacts when the listener rotated their head.

F. Source Position Smoothing. To address perceptual instability during head motion, exponential smoothing was applied to spatial parameters. Each source maintains both a current and target azimuth and elevation. Position updates follow

$$\theta_t = \alpha\theta_{t-1} + (1-\alpha)\theta_{\text{target}},$$

where α is derived from a desired transition time and audio buffer update rate. This smoothing significantly reduced rapid angular jumps that previously produced audible crackling.

G. HRTF Coefficient Interpolation and Crackling Mitigation. Early versions of the spatial audio pipeline exhibited audible crackling artifacts during head or source motion. Through systematic debugging, this issue was traced to abrupt transitions between discrete head-related impulse responses (HRIRs). Because HRTFs are measured at a finite set of azimuth and elevation angles, naïvely updating the spatial filter by selecting the nearest HRIR caused sudden, discontinuous changes in the convolution kernel. These discontinuities manifested perceptually as crackling or zipper noise.

Two complementary strategies were implemented to address this issue: spatial interpolation between neighboring HRIRs and temporal smoothing of HRIR coefficients across frames.

G.1. Spatial Interpolation Between HRIR Directions. Rather than selecting a single nearest-neighbor HRIR, the final system interpolates between multiple HRIRs corresponding to nearby directions on the unit sphere. Given a desired source direction $\mathbf{d}$ and a set of N nearest measured directions $\{\mathbf{d}_i\}$ with corresponding HRIRs $h_i[n]$, a weighted interpolation is performed:

$$\hat{h}[n] = \sum_{i=1}^{N} w_i \, h_i[n],$$

where the weights w_i satisfy

$$\sum_{i=1}^{N} w_i = 1, \quad w_i \geq 0.$$

Weights are computed as a function of angular distance between $\mathbf{d}$ and $\mathbf{d}_i$, such that closer HRIR measurements contribute more strongly. In practice, inverse-distance weighting is used:

$$w_i = \frac{1/\|\mathbf{d} - \mathbf{d}_i\|}{\sum_{j=1}^{N} 1/\|\mathbf{d} - \mathbf{d}_j\|}.$$

This spatial interpolation ensures that small changes in head orientation produce smooth changes in the effective impulse response, eliminating hard filter switches.

G.2. Temporal Smoothing of HRIR Coefficients. Although spatial interpolation reduced large discontinuities, residual crackling remained during rapid head motion due to frame-to-frame variation in interpolated HRIRs. To address this, exponential smoothing was applied directly to the HRIR coefficients over time.

Let $\hat{h}_t[n]$ denote the interpolated HRIR at audio frame t. The smoothed HRIR $h_t[n]$ is computed as

$$h_t[n] = \alpha h_{t-1}[n] + (1 - \alpha)\hat{h}_t[n],$$

where $\alpha \in [0, 1)$ is a smoothing coefficient determined empirically based on the audio buffer rate and desired responsiveness. Larger values of α increase temporal stability at the cost of responsiveness, while smaller values prioritize rapid spatial updates.

This temporal filtering effectively enforces continuity in the convolution kernel, preventing abrupt spectral changes that would otherwise generate audible artifacts.

G.3. Combined Effect on Audio Quality. The combination of spatial HRIR interpolation and temporal coefficient smoothing eliminated perceptual crackling while preserving directional cues and responsiveness. Importantly, these techniques operate directly on the impulse response level rather than on audio output, ensuring that stability is enforced at the source of the artifact. This approach proved more effective than increasing buffer sizes or relying solely on output-domain smoothing, and forms a core component of the final spatial audio pipeline.

H. Distance Attenuation and Amplitude Control. Distance-based attenuation is applied using a bounded inverse-distance model with conservative gain limits to prevent excessive near-field amplification. Additional global attenuation is applied to maintain consistent listening levels and avoid clipping when multiple sources are active.

Table 1. Summary of latency and jitter measurements for baseline and full AAR systems.

System	Mean (ms)	Std. Dev. (ms)	Notes
Baseline Visualizer	-256	339.217	High temporal variance
Full AAR Audio	-429.284	-10.579	Stable but buffered

I. Head-Tracking Integration. Head orientation updates are mapped to audio azimuth using a world-locked reference frame. Yaw rotation is inverted such that turning the head left causes the audio scene to rotate right, preserving spatial stability. Pitch updates are optionally applied with reduced gain to avoid excessive vertical motion artifacts.

Orientation updates are applied asynchronously within the audio callback loop to prevent blocking behavior and audible dropouts.

3. Results to Date

Initial system validation focused on timing behavior, stability, and perceptual responsiveness rather than absolute localization accuracy. The integrated system demonstrates stable head-tracked spatial audio with perceptually smooth transitions during head motion.

Latency analysis revealed that unsynchronized clocks between the IMU platform and host machine produced negative latency artifacts. As a result, evaluation focused on relative latency and jitter, which remain meaningful under clock offset.

4. Challenges and Lessons Learned

Several technical challenges were encountered during development. Unsynchronized system clocks produced physically impossible negative latency measurements, highlighting the importance of clock alignment in distributed real-time systems. Additional challenges included coordinate frame misalignment between the IMU and spatial audio reference frames, perceptual attenuation due to excessive smoothing, and yaw reference ambiguity inherent to 6-DOF inertial sensing without a magnetometer. Addressing these challenges required iterative debugging and careful tradeoffs between responsiveness and perceptual stability.

5. Next Steps

Planned work for Spring 2026 includes integration of computer vision for object detection, exploration of aesthetic soundscape techniques for representing object shape and size, migration to embedded wearable hardware, and preparation for a user study with blind and low-vision participants.

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References